

Classifying Extragalactic vs. Stellar Sources in the Chandra Source Catalog

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Motivation

The Chandra Source Catalog (CSC 2.1) contains tens of thousands of X-ray point sources, most without physical classification. X-ray emission arises from a wide range of objects including stellar coronae, compact binaries, and AGN, and distinguishing these populations matters for survey science and follow-up prioritization. This project builds a binary classifier to separate stellar sources from AGN using only catalog photometry and spectral features, without requiring optical spectra or manual review.

Data Source

Data was drawn from two sources. CSC 2.1 was queried via its public TAP service using `pyvo`, retrieving sources by detection significance ($>3\sigma$) in two dataset sizes: 5,000 and 20,000 sources. Features collected include sky coordinates, hardness ratios in three bands (HR_HS, HR_HM, HR_MS), and aperture fluxes in broad, soft, medium, and hard bands. Galactic coordinates were computed with `astropy`. Ground-truth labels came from cross-matching each source against SIMBAD at a 3 arcsecond radius using `astroquery.simbad`. SIMBAD object type codes were mapped to three categories: AGN (Seyferts, quasars, blazars, galaxies), stellar (stars, compact objects, X-ray binaries), and other. Sources with no match were left unlabeled. The full data collection pipeline is implemented in `query.py`, which saves the enriched catalog to CSV.

Data Analysis

Exploratory analysis and hypothesis testing were conducted in `dsa210_eda_hypothesis.ipynb` using `pandas`, `numpy`, `matplotlib`, and `scipy`. A sky plot showed AGN distributed isotropically while stellar sources trace the Galactic plane. AGN have roughly 8x higher median broadband flux than stellar sources, and tighter hardness ratio distributions, consistent with their more uniform spectral properties.

I tested whether AGN have harder X-ray spectra than stellar sources. The null hypothesis was that AGN and stellar hardness distributions are identical; the alternative (one-sided) was that AGN are stochastically greater. Both HR_HS and HR_HM were tested with Mann-Whitney U tests via `scipy.stats.mannwhitneyu`. Both tests rejected the null hypothesis: HR_HS yielded $p = 8.05e-3$ and HR_HM yielded $p = 3.70e-17$. The separation is much stronger in HR_HM, indicating that the hard/medium band is the more reliable spectral discriminator between the two classes.

Feature engineering and model training are implemented in `agn_stellar_classifier_5k.py` and `agn_stellar_classifier_20k.py`. Two features were engineered: $\log_{10}(\text{flux_aper_b})$ to compress the six-decade flux range, and $|\text{galactic latitude}|$ to capture the AGN/stellar spatial separation. A Random Forest (400 trees, balanced class weights, median imputation) was built using `scikit-learn`'s `RandomForestClassifier` inside a `Pipeline` with `SimpleImputer`, trained on labeled sources with a 75/25 stratified split, then applied to unlabeled and "other" sources to produce probability estimates saved to CSV.

Findings

Both models performed well, with the 20k model outperforming the 5k model across all metrics, reflecting the benefit of more training data. The 5k model (2,470 labeled sources) achieved 88.8% accuracy and ROC AUC of 0.959, while the 20k model (8,512 labeled sources) achieved 91.5% accuracy and ROC AUC of 0.964. Precision and recall were more balanced across both classes in the 20k model, whereas the 5k model showed a larger gap between stellar precision (0.925) and AGN precision (0.848).

In both models, $|\text{galactic latitude}|$ was by far the most important feature (importance 0.477 in 5k, 0.544 in 20k), followed by the hardness ratios. Log flux ranked lower than expected, suggesting that spatial position relative to the Galactic plane is the dominant discriminator. The hypothesis tests confirmed that AGN spectra are statistically harder than stellar spectra, particularly in the hard/medium band, consistent with the hardness ratios contributing meaningfully to classification.

When applied to unlabeled sources, the two models gave noticeably different AGN fractions: 39.3% (5k) vs. 58.0% (20k). This discrepancy likely reflects the fact that the 20k dataset reaches fainter, more distant sources, which are predominantly extragalactic.

Limitations and Future Work

SIMBAD cross-matching at 3 arcseconds introduces potential labeling noise from chance coincidences. Training labels also reflect SIMBAD's coverage bias toward bright, well-known sources. The binary framing discards the "other" class (clusters, nebulae) even though some of those sources may be misclassified.

Future work could incorporate variability and extent flags from CSC 2.1, expand to multi-class classification, and use cross-validation for more robust evaluation. Refining the SIMBAD type mapping using the full object-type hierarchy could also improve label quality.